EVERY Maths question (portal, till now)

Generated: 2026-10-24T00:20:29 by `scrape.py --export Maths`

For the AI tutor: teach these in due-date order. Status and XP are the student's real portal state. Question text is included in a code block wherever the tracker captured it; questions without text are listed at the end.

Official Math-1 syllabus (Slack #academic-announcement-26, read-only harvest 2026-10-23)

Math-1 Contest 1 — held 18 Sep 2026, 9:00-11:00 AM

Course Syllabus: Chapters 1–3

Chapter 1: Algebraic Expressions

Chapter 2: Quadratic Equations

Chapter 3: Inequalities

Math-1 Contest 2 — Fri 23 Oct 2026, 9:00-11:30 AM (Platform: Product)

Math-1 Contest 2 Syllabus

Chapter 1: Algebraic Expressions

Chapter 2: Quadratic Equations

Chapter 3: Inequalities

Chapter 4: Logarithms and Exponents

Chapter 5: Divisibility, GCD/HCF, and LCM

Chapter 6: Congruence, Modular Arithmetic, and the Euclidean Algorithm

Problem-Solving Sheets: 1 and 2

Math-1 IA-1 (written paper, from 19 Oct announcement)

Chapter 1: Algebraic Expressions

Chapter 2: Quadratic Equations

Chapter 3: Inequalities

Chapter 4: Logarithms and Exponents

Chapter 5: Divisibility, GCD/HCF, and LCM

Chapter 6: Congruence, Modular Arithmetic, and the Euclidean Algorithm

Problem-Solving Sheets: 1 and 2

Course textbooks (shared #m1-26, 2 Sep; download from Slack — files need your Slack login, so not mirrored here)

- challenges and thrills of pre college mathematics
- discrete mathematics and its applications 8e kenneth rosen 9781259676512

- precalculus mathematics by james stewart
- david burton - elementary number theory

Math-1 Contest 1 question paper (shared #m1-26, 25 Sep: M1_Contest_1_S_VYASA.pdf; full text extracted below — original needs Slack login)

Original: https://files.slack.com/files-pri/T0BNJRD9QT1-F0C40DNKB7Z/download/m1_contest_1_s_vyasa.pdf

Swami Vivek¯ ananda Yoga Anusandh¯ ana

Sa ˙ msth¯ ana

(S-VYASA)

(Deemed to be University)

USN

Examination M1 Contest Academic Y ear 2026–2027

Programme B.Tech Specialization AI

Semester I Date of Examination 18/09/2026

Course Code Session

Course Name Mathematics for Problem Solving

Name: Programme/Section:

Maximum Duration:90 minutesMaximum Marks:35

Instructions:

1. All questions are compulsory. Assume all variables are real unless stated otherwise.

2. In Section A, select one correct option; in Section B, write the required value or expression.

3. In Section C, match each statement with the correct numbered expression.

4. Show clear reasoning and working in Section D.

Section A: Multiple-Choice Questions 5 Q×1 M = 5 M

Que.

No. Questions CO RBTL Marks

1 A company sells a video game. If its price isprupees, the company estimates that

it will sell 20,000-500pgames. Which expression represents the revenue?

A)(20,000-500p) +pB)(20,000-500p)-p

C) 20,000-500pp D)p(20,000-500p)

CO1 2 1

2 The linesy=xandy= 3xpass through the origin. A student claims that the

slope of a line bisecting the angle between them is (1 + 3)/2 = 2. Why is this

reasoning incorrect?

A)Slopes should be multiplied rather than averaged.

B)An angle bisector averages the angles made with thex-axis, not their slopes;

slope is the tangent of the angle.

C)The correct slope must be

√

3, the geometric mean of the slopes.

D)A line through the origin cannot have the formy=mx.

CO1 3 1

3 Letp(x) =x 2 +ax+bhave rootscandd. Define

q(x) =x 2 + (2c+a)x+c 2 +ac+b.

Which of the following are roots of q(x)?

A)Its roots are 0 andd−c.B)Its roots arecandd+c.

C)Its roots arec 2 andd 2.D)It has the same roots asp(x).

CO2 3 1

Que.

No. Questions CO RBTL Marks

4 Ifa < b, which statement is always true?

A)−2a <−2bB)a+ 7< b+ 7

C)a 2 < b2 D) 1a > 1b

CO3 2 1

5 The graph ofy=x 2 −4x+ktouches thex−axis at exactly one point. Which

condition must hold?

A)The discriminant is negative.B)The discriminant is zero.

C)k= 0.D)The roots have opposite signs.

CO2 2 1

Section B: Fill in the Blanks 5 Q×1 M = 5 M

Que.

No. Questions CO RBTL Marks

6 A falling object travelsd= 16t 2 feet aftertseconds. After 6 seconds, it has

travelled feet.

CO2 2 1

7 Aarav and Meera have positive integer agesxandy, and both are at least 3 years

old. The cluesx+y= 12 and “two years ago, the sum of their ages was 8” allow

different ordered pairs (x, y).

CO1 3 1

8 A square has side lengthx. Four 1×1 corner squares are removed. The area of

the remaining region is .

CO1 2 1

9 If 1x >0, thenxis . CO3 2 1

10 The equationx 2 + (k−3)x+k= 0 has equal roots. The sum of all possible values

ofkis .

CO2 3 1

Section C: Match the Following 1 Q×5 M = 5 M

Que.

No. Questions CO RBTL Marks

11 A student club sends a fundraising appeal tonstudents and sets a target ofT

rupees. Match each description in Column A with its expression in Column B.

Column A Column B

A.Amount raised on reaching half the

target

1.T+ 500

B.Amount raised if every student gives

Rs. 50

2. T2

C.Amount raised if the club exceeds its

target by Rs. 500

3.25n

D.Amount still required after every

student gives Rs. 50

4.50n

E.Amount raised if half the students

give Rs. 50

5.T−50n

CO1 2 5

Section D: Attempt the Following Questions 5 Q×4 M = 20 M

Que.

No. Questions CO RBTL Marks

12 Saira has the following pieces:

•one square of sidexunits;

•eight rectangular strips, each measuringx×1 units; and

•fifteen squares, each of side 1 unit.

She claims that all the pieces can be arranged, without gaps or overlaps, to form

one large rectangle.

CO1 3 4

(i)Write an algebraic expression for the total area of all the pieces. (2)

(ii)Use the expression to determine the length and breadth of the resulting

rectangle in terms ofx.

(2)

13 The first three steps of a growing square pattern contain 1, 4 and 9 small squares

respectively.

Step 1 Step 2 Step 3

CO1 3 4

(i)Write an equation relating the step numbernand the number of squaresy,

and explain how it represents the pattern.

(2)

(ii)Determine the numbers of small squares in Steps 4 and 15. (2)

3/4

Que.

No. Questions CO RBTL Marks

14 Intelligence Quotient is calculated using

IQ = Mental AgeChronological Age ×100.

CO3 3 4

(i)For children of chronological age 15 years whose IQ satisfies 90≤IQ≤150,

determine the possible range of mental age.

(2)

(ii)For children of chronological age 12 years whose mental age satisfies

9≤MA≤15, determine the possible IQ range.

(2)

15 A model rocket is launched from a platform. Its height above the ground aftert

seconds is

h(t) =−5t 2 + 20t+ 25.

CO2 3 4

(i)Find the zeros ofh(t) and interpret the physically meaningful zero. (2)

(ii)Find the maximum height without using calculus. (1)

(iii)A student calculatesh(6) =−35 and claims the rocket is 35 m underground.

Is this interpretation correct? Explain.

(1)

16 At the Word Store, consonants are free, while each vowelA, E, I, O, Uhas its own

fixed price. The following word prices are known:

Word BANANA MATRIX CUBIC PENCIL ROBOT

Cost (Rs.) 9 7 9 10 4

Determine the cost of the wordEDUCA TION. Your solution must explain how

the given word prices determine the required vowel prices; simply stating the final

cost is not sufficient.

CO1 4 4

4/4

Lecture Worksheet 6 — Congruence and Modular Arithmetic (shared #m1-26, 5 Oct; 15 pages, full text extracted below — original needs Slack login)

Original: https://files.slack.com/files-pri/T0BNJRD9QT1-F0C6KN3T81H/download/l_6.pdf

Newton School of Technology Mathematics-1

Congruence and Modular Arithmetic

Worksheet 6

Student Name: Student ID:

This lecture will help you understand

•quotient, remainder, and the meaning of modulo,

•standard remainders and negative representatives,

•cyclic patterns in modular arithmetic,

•remainder families and congruence modulon,

•properties of congruence and reduction of large calculations.

“I see, I forget.

I hear, I remember.

I do, I understand.”

– Confucius

Newton School of Technology Mathematics-1

Quick Revision: GCD

1. Two numbers have

gcd(a, b) = 12,lcm(a, b) = 420.

If one number is60, find the other.

Answer:

2. Use the Euclidean Algorithm to findgcd(252,198).

Complete:

252 = 198( ) +

198 = ( ) +

= ( ) +

∴gcd(252,198) =

The Machine in the Basement

In this worksheet, we will explore division and remainders through a conversation between Ant and Gnam. As they

interact with the strange machine, follow their questions and complete the tasks along the way.

Warm-up Problem!

One evening, Ant and Gnam find an old locked basement under the Mathematics Lab.

Inside the basement, they see a strange machine. On the machine, something is written:

“Feed me a number. I will tell you what is left behind.”

Gnam:“What does that mean?”

Ant:“It sounds like a remainder machine.”

Gnam:“Remainder? Like division?”

Ant:“Exactly. But this machine will take us much deeper than ordinary division.”

The machine suddenly opens a small tray. Inside it are chocolates and empty boxes.

Machine:“Each box holds exactly5chocolates. Put17chocolates inside me.”

Gnam puts17chocolates into the machine. The machine asks them to complete its first message:

17 = 5× +

Full boxes= Leftover chocolates=

The machine makes a soft sound and prints more number-trials. Ant says, “Let us record the full boxes and the leftovers

each time.”

Activity 1: Quotient and Remainder

Number Divisor Quotient Remainder Division F orm

17 5 17 = 5( ) +

23 7 23 = 7( ) +

36 8 36 = 8( ) +

108 3 108 = 3( ) +

129 7 129 = 7( ) +

After looking at the machine’s answers, Ant asks:

What do you notice about the remainder in each case?

Y our Answer:

After the warm-up, let us make our understanding of division more precise by recalling!

Divide each number by5and write the remainder.

5÷5⇒Remainder=

6÷5⇒Remainder=

7÷5⇒Remainder=

8÷5⇒Remainder=

9÷5⇒Remainder=

So, when dividing by5, the possible remainders are :

, , , , .

Can we get any other possible remainder when dividing by5?

□Yes□No

Therefore, when dividing by5, the remainders go from

0,1, ...

Now think generally.

When a number is divided byn, the possible remainders are

0,1,2, . . . , .

These become thenpositions of our cycle.

Scene 1: The Circular Track

The machine lights up. A circular track appears on the floor with five positions:

0,1,2,3,4.

After4, the track returns to0.

0

1

23

4

Gnam asks:“Why are there only five positions? Where did the other numbers go?”

Ant explains:“They did not disappear. They are walking around the same cycle again and again. This is the idea of

workingmodulo5: divide by5, ignore the complete rounds, and keep only the final position or remainder.”

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Definition

Modulonmeans remainders after division byn. Equivalently, it means working on a cycle withnpositions:

0,1,2, . . . , n−1.

So modulo5means we only care about one of the five positions0,1,2,3,4.

Starting from0, Gnam walks17steps around the track.

17 = 5(3) + 2.

After every5steps, he comes back to0. After15steps, he is back at0, and2more steps take him to position2.

So, when Gnam walks17steps, he lands on position2.

Now try a few more walks and check where Gnam lands each time.

Use the cycle0→1→2→3→4→0.

Complete the table.

Steps Division Form Landing Position

7 7 = 5( ) +

12 12 = 5( ) +

18 18 = 5( ) +

23 23 = 5( ) +

31 31 = 5( ) +

What do you notice?

The landing position is the same as the .

Scene 2: The Strange Negative Walk

Gnam now walks backwards on the same modulo5cycle.

Starting at0, complete the landing positions.

1) Walking1step backward from0lands on .

2) Walking2steps backward from0lands on .

3) Walking3steps backward from0lands on .

4) Walking3steps backward reaches the same position as walking steps forward.

Gnam asks:“Can walking backward also describe a position on the cycle?”

Ant says:“Let us check this with a number.”

Now look at17.

First write17using the usual division form:

17 = 5( ) +

So the landing position of17on the modulo5cycle is .

Now look at the next multiple of5after17.

20 = 5( )

How many steps backward from20gives17?

4

So we can also write

17 = 5( )−

Now compare the two completed forms:

17 = 5(3) + 2and17 = 5(4)−3.

1) Do+2and−3give the same landing position on the modulo5cycle?

□Yes□No

2) If we describe the landing position by walking forward from0, which number do we use?

□2□−3

3) In division language, this forward landing position is called the standard remainder.

So the standard remainder is .

4) But −3is also useful. Instead of saying “2steps forward,” it describes the same landing position as “3steps

.”

Gnam says:“Let me check if I understood this game on another cycle.”

Ant says:“Good. Try the modulo7cycle now.”

1) On the modulo7cycle,10lands at the same position as .

2) On the modulo7cycle,−3lands at the same position as .

3) On the modulo7cycle,−21lands at the same position as .

4) On the modulo7cycle,−23lands at the same position as .

5) On the modulo7cycle,25lands at the same position as .

6) On the modulo7cycle,−15lands at the same position as .

Ant says:“Good. You now know that different numbers can land at the same position on a cycle. Let us see how a

whole city uses this idea.”

Scene 3: The Gate of Titan

The machine opens a portal. Ant and Gnam enter Titan, a city where arithmetic wraps around.

At the gate, a guard says:

5

“Only citizens with remainder badge3modulo11may enter.”

Gnam shows his card:25. The guard allows him inside.

Gnam asks:“But my card says25, not3!”

The guard replies:“On Titan, numbers that leave the same remainder are treated as members of the same family.”

Ant asks:“The gate wants remainder badge3modulo11. Can more than one number have this badge?”

□Yes□No

Gnam says:“Let me check by moving full cycles of11.”

3−22 = ,3−11 = ,3 = ,3 + 11 = ,3 + 22 = .

Do all these numbers land at badge3modulo11?

□Yes□No

Look at the numbers above carefully.

What pattern do you observe?

Using this pattern, write the family of badge3modulo11.

{. . . , , ,3, , , . . .}

Ant asks:“Instead of listing forever, can you write the whole family using one expression?”

Family of3modulo11 ={ : }.

Gnam says:“Now I think I can write this for any badge.”

If the badge isaand the cycle length isn, first write the nearby family members:

{. . . , , , a, , , . . .}

Now write the full family in one expression.

Family ofamodulon={ : }.

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Scene 4: Gnam’s Difference Test

Gnam says:“Numbers in the same family look different, but maybe their difference tells us something.”

Ant says:“Let us test a few pairs first.”

The following pairs belong to the same family modulo11. Complete the table.

Pair Difference Divisible by11?

25,3 25−3 = □Yes□No

47,14 47−14 = □Yes□No

36,25 36−25 = □Yes□No

−8,3 −8−3 = □Yes□No

What do you notice about the differences?

Your Answer:

Ant asks:“Can we explain why this always happens?”

One member of the family of3modulo11can be written as

3 + 11p.

Write another member of the same family:

Now subtract the two members and simplify.

(3 + 11p)−

What number is always a factor in the difference?

Answer:

So, if two numbers belong to the same family modulo11, their difference is divisible by .

Ant finally introduces the symbol for“same cycle position.”Since25and3leave the same remainder when divided

by11, mathematicians write

25≡3 (mod 11).

This is read as:“25 is congruent to 3 modulo 11.”

Ant says:“We have seen the idea through families and differences. Now let us write the same idea formally.”

What doesa≡b(modn)mean?

Leta, b∈Zand letnbe a positive integer. We say that

a≡b(modn)

ifaandbleave the same remainder when divided byn. Equivalently,

a≡b(modn)⇐ ⇒n|(a−b).

Equivalently,a=b+knfor some integerk.

Scene 5: The Modular Addition Game

Ant and Gnam enter Titan’s playground.

For this game, only totals that appear after a player’s turn are counted as previous totals.

1. Player A chooses a positive integern.

2. Player B decides who goes first.

3. The running totalTstarts at0.

4. On each turn, a player chooses any integermand adds it toT.

5. A player loses if the new total is congruent to any previous total modulon.

Understand the Game

Alice choosesn= 5. Bob goes first.

Round 1 2 3 4 5

Alice −88 −23

Bob 7 102 13

Running Total

T

7 −81 21 −2 11

Remainder

mod5

Bob loses because11≡21 (mod 5). What number could Bob have picked instead of13?

Your Answer:

Play the Game

Player A: Player B: Modulusn:

Round 1 2 3 4 5 6 7 8 9 10

Player A

Player B

TotalT

Remainder

How many safe totals can appear before a remainder must repeat?

Your Answer:

Ant says:“In the game, only the final landing position mattered. Now let us see whether we can replace large

numbers by smaller numbers before calculating.”

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Scene 6: Reduce First, Then Calculate

Gnam says:“These numbers may look big, but maybe we do not need to carry the full numbers.”

Ant says:“Let us check whether smaller representatives give the same landing position.”

Work modulo11.

First reduce the numbers.

13≡ (mod 11) 27≡ (mod 11)

Now check addition in two ways.

13 + 27 = , ≡ (mod 11)

2 + 5 = ,7≡ (mod 11)

So,

13 + 27≡ + (mod 11).

Leta, b, c, d∈Zand letnis a positive integer.

If

a≡b(modn)andc≡d(modn),

then

a+c≡b+d(modn).

Now, check subtraction in two ways.

27−13 = , ≡ (mod 11)

5−2 , ≡ (mod 11)

So,

27−13≡ − (mod 11).

Leta, b, c, d∈Zand letnis a positive integer.

If

a≡b(modn)andc≡d(modn),

then

a−c≡b−d(modn).

Now, check multiplication in two ways.

13·27 = , ≡ (mod 11)

2·5 = , ≡ (mod 11)

So,

13·27≡ · (mod 11).

Leta, b, c, d∈Zand letnis a positive integer.

If

a≡b(modn)andc≡d(modn),

then

ac≡bd(modn).

Now multiply both sides by the same number.

13≡ (mod 11)

Multiply both sides by4.

4·13 = , ≡ (mod 11)

4·2 = , ≡ (mod 11)

So,

4·13≡ · (mod 11).

Leta, b∈Zand letnis a positive integer.

If

a≡b(modn),

then for every integerk,

ka≡kb(modn).

Now, raise both sides to the same power.

13≡ (mod 11)

Square both sides.

132 = , ≡ (mod 11)

22 = , ≡ (mod 11)

So,

132 ≡ (mod 11).

Leta, b∈Zand letnis a positive integer.

If

a≡b(modn),

then for every positive integerk,

ak ≡b k (modn).

Now, check a chain of same-family numbers.

24≡ (mod 11) 13≡ (mod 11)

So,

24≡13 (mod 11)and13≡2 (mod 11).

Therefore,

24≡ (mod 11).

Leta, b∈Zand letnis a positive integer.

If

a≡b(modn)andb≡c(modn),

then

a≡c(modn).

Now at another gate of Titan, the cycle length is6. The guard shows Gnam a strange equality.

2·1 = 2·4 =

Now check their landing positions on the modulo6cycle.

2·1≡ (mod 6) 2·4≡ (mod 6)

So we get

2·1≡2·4 (mod 6).

Gnam asks:“Since both sides have2, can I cancel it?”

Can we write this?

1≡4 (mod 6)

□Yes□No

Check the landing positions of1and4on the modulo6cycle.

1≡ (mod 6) 4≡ (mod 6)

Are they the same?

□Yes□No

So, can we always cancel a common factor in congruences?

□Yes□No

Note

General Conclusion.

Leta, b, k∈Zandnis a positive integer, wherenis the cycle length. Letd= gcd(k, n). Then

ka≡kb(modn)⇐ ⇒a≡b(modn/d).

In particular, ifgcd(k, n) = 1, then

ka≡kb(modn) =⇒a≡b(modn).

Thus, a common factor can be cancelled without changing the modulus when it is coprime to the modulus.

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Practice Problems

Question 1.Find the remainder of46 100 when divided by5.

Your Answer:

Question 2.A number is calledthreevenif it leaves remainder0when divided by3,throdd1if it leaves remainder

1, andthrodd2if it leaves remainder2.

Complete the following modulo3, and then name the type of the answer.

threeven+throdd1≡ (mod 3)⇒threeven+throdd1 is of type

threeven+throdd2≡ (mod 3)⇒threeven+throdd2 is of type

throdd1−throdd2≡ (mod 3)⇒throdd1−throdd2 is of type

throdd2−throdd1≡ (mod 3)⇒throdd2−throdd1 is of type

threeven×throdd1≡ (mod 3)⇒threeven×throdd1 is of type

throdd1×throdd2≡ (mod 3)⇒throdd1×throdd2 is of type

Explain one rule out of all.

Your Answer:

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Question 3.Mary enters a sleep chamber at6pm. The chamber has a12-hour clock. She sets the timer for76hours.

At what time will the alarm ring?

Your Answer:

Question 4.Sam starts eating green eggs and ham on Monday. Consider Monday as Day1. He continues eating

them for200days.

On which day of the week will he stop?

Your Answer:

Question 5.A6 × 6chessboard is coloured black and white in the usual alternating pattern. Two opposite corner

squares are removed.

Can the remaining34squares be covered exactly by17dominoes, where each domino covers two adjacent squares?

Justify your answer.

Your Answer:

Question 6.Joanna starts by facingwest. She rotates counterclockwise by7110◦.

Cycle length for angles=

After reducing7110 ◦ using this cycle length, what direction will Joanna face? Justify your answer.

Your Answer:

A Frightful Halloween

Question 7.On a frightful Halloween night, Todd is waiting outside a haunted house. His friends count the ghosts,

but they give him only remainder clues.

Eileen counted the ghosts by3’s and found2left over.

Jane counted the ghosts by5’s and found3left over.

Eric counted the ghosts by19’s and found7left over.

Letxbe the number of ghosts. Then

$x \equiv 2 \pmod 3$, $x \equiv 3 \pmod 5$, $x \equiv 7 \pmod{19}$.

What is the smallest possible number of ghosts?

Your Answer:

Question 8.Some soldiers are standing in a group.

Letxbe the total number of soldiers.

```
When they line up in rows of5,2soldiers are left over.

When they line up in rows of7,4soldiers are left over.

The total number of soldiers is less than100.

Find all possible values ofx.

Your Answer:

14

Newton School of Technology Mathematics-1

Gnam:“So remainder is not just what is left after division. It is also a position in a cycle.”

Ant:“Yes. Modular arithmetic is the mathematics of cycles.”

Gnam:“And now big numbers do not look so big anymore.”

Why Should an Engineer Care?

Modular arithmetic is one of the hidden tools behind modern computing and Artificial Intelligence.

• Cryptography:Secure communication on the internet (RSA, digital signatures, blockchain) relies
heavily on

modular arithmetic.

• Computer Arithmetic:Computers store numbers using a fixed number of bits. Overflow behaves like

arithmetic modulo2 n.

• Error Detection and Coding:QR codes, communication systems, and data storage use modular
arithmetic

to detect and correct errors.

• Machine Learning Systems:Large-scale AI software relies on efficient numerical computation, where

modular ideas frequently appear in optimization, hashing, indexing, and distributed systems.

In short:whenever a system wraps around, repeats in cycles, encrypts information, hashes data, or
performs

efficient large-number computations, modular arithmetic is working behind the scenes.

End of Worksheet

15
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Contest 2 hint image (shared #m1-26, today: hint for last question of today's contest)

https://files.slack.com/files-pri/T0BNJRD9QT1-F0CBDQ1SPHB/download/23_oct.png

Maths 1 Learning Tracker (Google Sheet, shared #academic-announcement-26 7 Sep / 18 Sep — view-only; File > Make a copy to tick your own progress) — 172

Topic rows (Lecture | Sub-topics | Concept check):

- Algebraic Equation: Constant and variable

- Algebraic Equation: Algebraic Expression
- Algebraic Equation: Difference Between Expression and Equation
- Algebraic Equation: Linear Equation
- Algebraic Equation: Linear Equation in 1 Variable
- Algebraic Equation: Solution of System of Linear Equations
- Algebraic Equation: Linear Equation in 2 Variables
- Algebraic Equation: Convert a Word Problem into an Equation
- Algebraic Equation: Methods to Solve Linear Equation in 2 Variables — Substitution, Elimination and Graphical Method
- Algebraic Equation: Solution of Linear Equation in 2 Variables — Unique, Infinite and No Solution
- Quadratic Equation: Standard Form of Quadratic Equation
- Quadratic Equation: Need of Quadratic Equation
- Quadratic Equation: Methods for Solving Quadratic Equation — Factorization and Completing the Square Method
- Quadratic Equation: Quadratic Formula
- Inequalities: Equality vs Inequality
- Inequalities: Uses and Applications of Inequality
- Inequalities: Linear Inequality — Linear Inequality in One and Two Variables
- Inequalities: Quadratic Inequality
- Problem Solving Session: Invariance Principle
- Problem Solving Session: Parity Principle
- Problem Solving Session: Extremal Principle
- Exponent and Logarithm: Multiplication and Addition World
- Exponent and Logarithm: Exponent as Repeated Multiplication
- Exponent and Logarithm: Laws of Exponents
- Exponent and Logarithm: Relationship Between Logarithm and Exponent
- Exponent and Logarithm: Laws of Logarithms
- Exponent and Logarithm: Change of Base
- Exponent and Logarithm: Applications of Logarithm and Exponent
- LCM and GCD: Division as a Structure
- LCM and GCD: Divisibility
- LCM and GCD: GCD and LCM
- LCM and GCD: Methods to Find GCD and LCM — Brute Force and Factorization Method
- LCM and GCD: Fundamental Theorem of Arithmetic
- LCM and GCD: Relation Between GCD and LCM
- LCM and GCD: Division Algorithm
- LCM and GCD: Euclidean Algorithm
- Modulo Arithmetic: Division and Modulo Difference — Quotient and Remainder
- Modulo Arithmetic: Standard Remainders
- Modulo Arithmetic: Cyclic Pattern in Modulo
- Modulo Arithmetic: Congruence Classes
- Modulo Arithmetic: Properties of Modulo Arithmetic
- Modulo Arithmetic: Computation in Modular Arithmetic
- Problem Solving Session: Invariance Principle
- Problem Solving Session: Parity Principle
- Problem Solving Session: Extremal Principle
- Lines: Cartesian Coordinate System
- Lines: Distance Between Two Points

- Lines: Midpoint of a Line Segment
- Lines: Slope and Its Interpretation — Slope as Rise Over Run
- Lines: — Slope as rate of change
- Lines: — Interpreting positive,negative, zero and undefined slope
- Lines: Equation of a Line — Slope-Intercept Form
- Lines: — Slope point form
- Lines: — Two point form
- Lines: Relationship Between Lines — Parallel, Perpendicular and Intersecting Lines
- Circle: Equation of a Circle — Standard and General Form
- Circle: Geometric Features of Circle — Identifying Centre and Radius from Equation
- Circle: Position of a Point Relative to a Circle — Points Inside, On and Outside the Circle
- Circle: Line-Circle Intersection — Secant and Tangent
- Circle: — Number of intersecting point
- Circle: Tangent to a Circle — Tangent at a Point on Circle
- Circle: — Radius tangent perpandicularity
- Circle: — Equation of a tangent
- Problem Solving Session: Invariance Principle
- Problem Solving Session: Parity Principle
- Problem Solving Session: Extremal Principle
- Sequence and Series: Pattern Recognition — Identifying Patterns
- Sequence and Series: — Explicit and recursive rule
- Sequence and Series: — finding the nth Term
- Sequence and Series: Arithmetic Progression — First Term, Common Difference and nth Term
- Sequence and Series: — Sum of the First n Terms
- Sequence and Series: — Applications
- Sequence and Series: Geometric Progression — First Term, Common Difference and nth Term
- Sequence and Series: — Sum of a Finite and Infinite Geometric Progression
- Sequence and Series: — Applications
- Sequence and Series: Sigma Notation — Introduction to Sigma Notation
- Sequence and Series: — Properties of Summation
- Sequence and Series: — Double Summations
- Sequence and Series: Telescoping Sums — Special Summation Techniques
- Sequence and Series: — Partial Fraction Decomposition
- Sequence and Series: — Cancellation Patterns
- Counting 1: Addition Principle — Counting Outcomes from Separate, Non-Overlapping Cases
- Counting 1: — Difference between “AND” and “OR” situations
- Counting 1: Multiplication Principle — Repetition Allowed
- Counting 1: — No repetition
- Counting 1: Permutations and Combinations
- Counting 1: Arrangements with Repeated Objects
- Counting 1: Complementary Counting
- Counting 1: Restricted Arrangements
- Counting 2: Inclusion-Exclusion Principle
- Counting 2: Pigeonhole Principle — Identifying Pigeons and Holes
- Counting 2: Generalised Pigeonhole Principle
- Counting 2: Recurrence Relations
- Counting 2: Recurrence Relations — Building a Large Case from Smaller Cases
- Counting 2: Double Counting — Counting the Same Collection in Two Ways

- Counting 2: Stars and Bars
- Problem Solving Session: Invariance Principle
- Problem Solving Session: Parity Principle
- Problem Solving Session: Extremal Principle
- Sets: Set Fundamentals — Builder and Roster Form
- Sets: Membership and Subset — Elements and Membership
- Sets: — Subsets
- Sets: Types of Sets — Singleton and Empty Set
- Sets: — Finite and Infinite Set
- Sets: — Universal and Complementary Set
- Sets: Set Operations — Union and Intersection
- Sets: — Difference and complement
- Sets: Venn Diagrams
- Sets: Set Counting — Cardinality
- Sets: — Inclusion-Exclsion Principal
- Relation and Function: Ordered Pair
- Relation and Function: Cartesian Product — Cartesian Product of Two Sets
- Relation and Function: — Cardinality of A x B
- Relation and Function: Relations — Relation as a Subset of A × B
- Relation and Function: — Domain and Range of a Relaion
- Relation and Function: — Relation on a set
- Relation and Function: Properties of Relation — Reflexive, Symmetric and Transitive Relations
- Relation and Function: Functions — Function as a Special Relation
- Relation and Function: — Domain,codomain and range
- Relation and Function: — Distinguishing function from general relation
- Relation and Function: Special Functions — Floor, Ceiling, Signum and Constant Functions
- Graph Sketching 1: Graphs of Functions — Purpose and Applications of Graphs
- Graph Sketching 1: — Definition of a Graph
- Graph Sketching 1: — Identifying Functions: The Vertical Line Test
- Graph Sketching 1: — Reading and Interpreting Graphs
- Graph Sketching 1: Parent Functions — Polynomial Function
- Graph Sketching 1: — Square root function
- Graph Sketching 1: — Reciprocal Function
- Graph Sketching 1: — Absolute value function
- Graph Sketching 1: — Greatest Integer function
- Graph Sketching 1: — Exponential and Logarithmic function
- Graph Sketching 2: Transformations of Functions — Purpose and Applications of Transformations
- Graph Sketching 2: — Vertical Shifts
- Graph Sketching 2: — Horizontal Shifts
- Graph Sketching 2: — Reflections of Graphs
- Graph Sketching 2: — Vertical Stretching and Compression
- Graph Sketching 2: — Horizontal Stretching and Compression
- Graph Sketching 2: — Absolute Value Transformations
- Problem Solving Session: Invariance Principle
- Problem Solving Session: Parity Principle
- Problem Solving Session: Extremal Principle
- Limit and Continuity: Intuitive Idea of a Limit — Approaching Values
- Limit and Continuity: Limits from Graphs

- Limit and Continuity: Limit Notation
- Limit and Continuity: Algebraic Evaluation of Limits — Direct Substitution
- Limit and Continuity: — Factorisation
- Limit and Continuity: — Rationalisation
- Limit and Continuity: One-Sided Limit — Left-Hand Limit
- Limit and Continuity: — Right-Hand Limit
- Limit and Continuity: Continuity at a Point — Existence of Limit
- Limit and Continuity: — Function Value Exists
- Limit and Continuity: — Limit Equals Function Value
- Derivative: Average Rate of Change
- Derivative: Instantaneous Rate of Change
- Derivative: Derivative from First Principles
- Derivative: Geometric and Physical Interpretation
- Derivative: Differentiability and Continuity
- Derivative: Basic Differentiation Rules — Power, Sum and Difference Rules
- Derivative: — Product and Quotient Rules
- Derivative: — Chain Rule
- Derivative: — Derivatives of Standard Functions
- Application of Derivative: Meaning and Sign of the Derivative
- Application of Derivative: Increasing and Decreasing Functions
- Application of Derivative: Critical and Stationary Points
- Application of Derivative: Local Maximum and Minimum
- Application of Derivative: First-Derivative Test
- Application of Derivative: Optimisation — Objective Function
- Application of Derivative: — Constraints
- Application of Derivative: — Maximum or Minimum Value
- Problem Solving Session: Invariance Principle
- Problem Solving Session: Parity Principle
- Problem Solving Session: Extremal Principle

#m1-26 — every message (13)

Akshit Sharma

m1-26

31 Aug at 7:10 PM

Hello Batch A,

This is a reminder that you have your Mathematics Lab and Lecture tomorrow. Please ensure that you arrive on time and be present in the classroom before the session begins. Late entry will not be permitted except in case of genuine operational issues.

Also, Kindly bring your respective problem solving notebook, as instructed earlier, as it will be required during the lab session.It will be checked.

Looking forward to seeing everyone prepared and on time. Let’s have a productive session tomorrow!

Regards,

Akshit Sharma (edited)

Akshay Telmasare

m1-26

2 Sep at 1:07 PM

Hello Students, Please find the 3 books we will be using as textbooks for this course in this semester. Download these books on your local machine. (edited)

Challenges and Thrills of Pre College Mathematics.pdf

Shared by Akshay Telmasare on 2 Sep

Discrete_Mathematics_and_Its_Applications_8e_Kenneth_Rosen_9781259676512.pdf

Shared by Akshay Telmasare on 2 Sep

PRECALCULUS MATHEMATICS BY JAMES STEWART.pdf

Shared by Akshay Telmasare on 2 Sep

LINKS: https://newtonschoolo-e998430.slack.com/files/U0BQJ0LCF2N/F0BUGKJHAQL/challenges_and_thrills_of_pre_college_mathematics.pdf?origin_team=T0BNJRD9QT1 https://files.slack.com/files-pri/T0BNJRD9QT1-F0BUGKJHAQL/download/challenges_and_thrills_of_pre_college_mathematics.pdf?origin_team=T0BNJRD9QT1 https://newtonschoolo-e998430.slack.com/files/U0BQJ0LCF2N/F0BUGLEBCKE/discrete_mathematics_and_its_applications_8e_kenneth_rosen_9781259676512.pdf?origin_team=T0BNJRD9QT1 https://files.slack.com/files-pri/T0BNJRD9QT1-F0BUGLEBCKE/download/discrete_mathematics_and_its_applications_8e_kenneth_rosen_9781259676512.pdf?origin_team=T0BNJRD9QT1 https://newtonschoolo-e998430.slack.com/files/U0BQJ0LCF2N/F0BUEU31DU1/precalculus_mathematics_by_james_stewart.pdf?origin_team=T0BNJRD9QT1 https://files.slack.com/files-pri/T0BNJRD9QT1-F0BUEU31DU1/download/precalculus_mathematics_by_james_stewart.pdf?origin_team=T0BNJRD9QT1

Akshit Sharma

m1-26

2 Sep at 9:45 PM

@channel

Dear Batch A Students,

Please bring Lecture Sheet 1, completed in full, along with your completed Lab Notebook.

Please ensure that both are completed and brought with you.

Thank you. (edited)

Akshit Sharma

m1-26

2 Sep at 9:46 PM

@channel

Dear Students,

Some students did not bring their Lecture Sheets today, while some students who were absent have not yet collected them.

Please ensure that you collect your Lecture Sheets and bring them to class, along with completing the required work.

Batch A & Batch B students, please make sure you have your Lecture Sheets with you.

Thank you. (edited)

Akshit Sharma

m1-26

7 Sep at 10:15 PM

@channel

Dear Batch A Students,

Please bring your Lab Sheet, Lecture Notebook, and Lab Notebook to class tomorrow.

As you know, we will be completing the chapter on Quadratic Equations in tomorrow’s session. Please revise the entire chapter beforehand and come prepared with any doubts you may have.

Make sure all your previous work is up to date and that you bring all the required materials.

Thank you.

Akshit Sharma

m1-26

10 Sep at 5:06 PM

@channel Hello Everyone please collect the sheets they are printed who were asking for lecture 1,2 or 3 or those who were absent.

Akshay Telmasare

m1-26

15 Sep at 6:02 PM

@channel

Dear Students, this is a gentle reminder for everyone that we are having our first Maths-1 contest on this Friday.

The syllabus for contest-1 is chapter 1-3 (meaning till Inequalities).

Kindly prepare accordingly. Best of Luck. (edited)

nikhil.menon

11 Sep

Posted in academic-announcement-26

@channel

REMINDER

Students,

On 18th September 2026, is your Math-1 Contest 1.

Contest Time: 9:00 AM to 11:00 AM

Reporting Time: 8:30 AM

Course Syllabus: Chapters 1–3

Chapter 1: Algebraic Expressions

Chapter 2: Quadratic Equations

Chapter 3: Inequalities

Show less

Akshay Telmasare

m1-26

25 Sep at 5:08 PM

Contest 1 Question Paper

M1_Contest_1_S_VYASA.pdf

Shared by Akshay Telmasare on 25 Sep

LINKS: https://newtonschoolo-e998430.slack.com/files/U0BQJ0LCF2N/F0C40DNKB7Z/m1_contest_1_s_vyasa.pdf?origin_team=T0BNJRD9QT1 https://files.slack.com/files-pri/T0BNJRD9QT1-F0C40DNKB7Z/download/m1_contest_1_s_vyasa.pdf?origin_team=T0BNJRD9QT1

Akshit Sharma

m1-26

5 Oct at 4:32 PM

L_6.pdf

Shared by Akshit Sharma on 5 Oct

LINKS: https://newtonschoolo-e998430.slack.com/files/U0BQJ0LCF2N/F0C6KN3T81H/l_6.pdf?origin_team=T0BNJRD9QT1 https://files.slack.com/files-pri/T0BNJRD9QT1-F0C6KN3T81H/download/l_6.pdf?origin_team=T0BNJRD9QT1

Akshit Sharma

m1-26

5 Oct at 6:04 PM

David_Burton_-_Elementary_number_theory.pdf

Shared by Akshit Sharma on 5 Oct

LINKS: https://newtonschoolo-e998430.slack.com/files/U0BQJ0LCF2N/F0C6M996AMB/david_burton_-_elementary_number_theory.pdf?origin_team=T0BNJRD9QT1 https://files.slack.com/files-pri/T0BNJRD9QT1-F0C6M996AMB/download/david_burton_-_elementary_number_theory.pdf?origin_team=T0BNJRD9QT1

Dhairya Agrawal

m1-26

19 Oct at 11:13 AM

Sir Pls Update What is the Syllabus for Maths Contest 2

Akshit Sharma

m1-26

19 Oct at 12:09 PM

Sending today

1

Akshay Telmasare

m1-26

Today at 11:15 AM

Hint for last question on today's contest.

1000064972.jpg

Shared by Akshay Telmasare today

LINKS: https://newtonschoolo-e998430.slack.com/files/U0BQJ0LCF2N/F0CB3441ZGX/1000064972.jpg?origin_team=T0BNJRD9QT1 https://files.slack.com/files-pri/T0BNJRD9QT1-F0CB3441ZGX/download/1000064972.jpg?origin_team=T0BNJRD9QT1

Akshit Sharma

m1-b1

10 Sep at 5:06 PM

@channel Hello Everyone please collect the sheets they are printed who were asking for lecture 1,2 or 3 or those who were absent.

Math-relevant messages from shared channels (#academic-announcement-26, #general-announcements-26, #student-progression-batch-a/b)

#academic-announcement-26

nikhil.menon

academic-announcement-26

31 Aug at 7:54 PM

@channel

Students,

Kindly join the respective Slack channels for each subject based on your assigned batch.

Please do not join channels of other batches. Make sure you join all subject channels corresponding to the batch you are enrolled in.

Kindly complete this at the earliest.

#m1-26 #m1-a1 #m1-a2 #m1-b1 #m1-b2

#snai-26 #snai-a1 #snai-a2 #snai-b1 #snai-b2

#psp-26 #psp-a1 #psp-a2 #psp-b1 #psp-b2

#physics-26 #physics-a1 #physics-a2 #physics-b1 #physics-b2

#academic-announcement-26

nikhil.menon

academic-announcement-26

3 Sep at 6:06 PM

@channel

Students,

Tomorrow's English class is cancelled for both batches A and B.

Revised LHL slots :

Batch A - 2:00 PM to 3:00 PM (Classroom 8)

Batch B - 3:00 PM to 4:00 PM (Classroom 8)

Batch B will have an EXTRA class for M1 from 4:00 PM to 5:00 PM. (Classroom 8)

#academic-announcement-26

nikhil.menon

academic-announcement-26

3 Sep at 10:56 PM

@channel

UPDATE

Students,

Revised LHL slots :

Batch A - 2:00 PM to 3:00 PM (Classroom 8)

Batch B - 3:15 PM to 4:15 PM (Classroom 8)

Batch B will have an EXTRA class for M1 from 4:15 PM to 5:15 PM. (Classroom 8) (edited)

#academic-announcement-26

nikhil.menon

academic-announcement-26

7 Sep at 3:20 PM

@channel

Applied Physics Syllabus for contest 1

Sizes in the Universe

Introduction to Measurement

Units and Dimensions

Order of Magnitude

Significant Digits

Scientific Notation

Error and Propagation

Scalars and Vectors

Vector Arithmetic

Vector Products

Vector Resolution

Differentiation

Integration

Distance and Displacement

Speed and Velocity

Acceleration and Jerk

Graphical Understanding of Motion

Motion in Straight Line

Equations of Motion

Relative Motion

Solving 1-D Motion

Projectile Motion

Breaking Down Multi-D Motion

i.e first four lectures (refer provided worksheets for reference and also practice worksheets questions), last 4 topics will be covered in your Wed's lecture be prepared with rest of the topics. (edited)

#academic-announcement-26

nikhil.menon

academic-announcement-26

10 Sep at 8:11 PM

@channel

Students,

Batch B will have an extra class for M1 tomorrow from 3:00 PM to 4:30PM in classroom 8.

Kindly take a note of the same. (edited)

#academic-announcement-26

nikhil.menon

academic-announcement-26

11 Sep at 2:12 PM

@channel

Students,

A mandatory Math lecture is scheduled for Batch B from 5:00 PM to 6:30 PM in classroom 8. (edited)

#academic-announcement-26

nikhil.menon

academic-announcement-26

11 Sep at 4:45 PM

@channel

REMINDER

Students,

On 18th September 2026, is your Math-1 Contest 1.

Contest Time: 9:00 AM to 11:00 AM

Reporting Time: 8:30 AM

Course Syllabus: Chapters 1–3

Chapter 1: Algebraic Expressions

Chapter 2: Quadratic Equations

Chapter 3: Inequalities

#academic-announcement-26

nikhil.menon

academic-announcement-26

17 Sep at 6:00 PM

@channel

REMINDER

Students,

Tomorrow, 18th September 2026, is your M1 Contest 1.

Contest Time: 9:00 AM to 11:00 AM

Reporting Time: 8:30 AM

Seating Arrangement: Will be shared by 8:00 AM tomorrow.

All students are requested to go through the exam guidelines carefully before the contest and ensure that you are familiar with all the instructions.

Please report on time and follow the exam guidelines strictly.

All the best!

Exam Guidelines for Student's.pdf

Shared by nikhil.menon on 11 Sep

LINKS: https://newtonschoolo-e998430.slack.com/files/U0BQJ0LCF2N/F0C107Y9AG3/exam_guidelines_for_student_s.pdf?origin_team=T0BNJRD9QT1 https://files.slack.com/files-pri/T0BNJRD9QT1-F0C107Y9AG3/download/exam_guidelines_for_student_s.pdf?origin_team=T0BNJRD9QT1

#academic-announcement-26

Payal Jain

academic-announcement-26

18 Sep at 5:45 PM

Sharing it again here:

@channel Here are the Learning trackers that covers topics across all four subjects:

S&AI: docs.google.com/spreadsheets/d/18zDsWPINbLh…/edit?usp=sharing

Mathematics 1: docs.google.com/spreadsheets/d/1pQz1BB1SDS8…/edit?gid=182933272#gid=182933272

Applied Physics: docs.google.com/spreadsheets/d/1ZOx2WQ11_7d…/edit?usp=sharing

PSP: docs.google.com/spreadsheets/d/1dSKXZJ_aOez…/edit?gid=1592514245#gid=1592514245

Once you've made a copy of all the trackers, please react to this message with a

.

6

LINKS: https://docs.google.com/spreadsheets/d/18zDsWPINbLhKLMVs8S13rtJQAFUp---_QMasaiwQqzI/edit?usp=sharing https://docs.google.com/spreadsheets/d/1pQz1BB1SDS8NPHzR5RHyG-Z_631Rb40ra2dl8aB0lxo/edit?gid=182933272#gid=182933272 https://docs.google.com/spreadsheets/d/1ZOx2WQ11_7dymZ7tux25VkwUmWyXea1pGLauEx-WNIU/edit?usp=sharing https://docs.google.com/spreadsheets/d/1dSKXZJ_aOezRekJ1RqTcPjQDa-AsB85xYugHMpaYYgA/edit?gid=1592514245#gid=1592514245

#academic-announcement-26

nikhil.menon

academic-announcement-26

22 Sep at 1:40 PM

@channel

Today's M1 and PSP labs are combined for Batch A.

M1 lab - 1:30 PM to 3:00 PM Classroom 6

PSP lab - 3:00 PM to 4:30 PM Classroom 6

#academic-announcement-26

nikhil.menon

academic-announcement-26

24 Sep at 6:08 PM

@channel

Students,

M1 Contest 1 paper distribution will be held tomorrow, as per the below schedule:

BATCH A: 2:00 PM to 3:00 PM

BATCH B: 5:00 PM to 6:00 PM

#academic-announcement-26

nikhil.menon

academic-announcement-26

25 Sep at 12:45 PM

@channel

REMINDER

Students,

M1 Contest 1 paper distribution will be held TODAY, as per the below schedule:

BATCH A: 2:00 PM to 3:00 PM (Classroom 6)

BATCH B: 5:00 PM to 6:00 PM (Classroom 8) (edited)

3 replies

Last reply 28 days ago

#academic-announcement-26

nikhil.menon

academic-announcement-26

25 Sep at 1:05 PM

@channel

REMINDER

Students,

M1 Contest 1 paper distribution will be held TODAY, as per the below schedule:

BATCH A: 2:00 PM to 3:00 PM (Classroom 6)

BATCH B: 5:00 PM to 6:00 PM (Classroom 8)

#academic-announcement-26

nikhil.menon

academic-announcement-26

28 Sep at 12:23 PM

@channel

Students,

Kindly come to Classroom 1 to check your math answer sheets at 6.15PM. This will the the last opportunity to check your sheets.

#academic-announcement-26

garv

academic-announcement-26

8 Oct at 8:55 AM

Hi guys, we need everyone’s input on how the classes are going for PSP, Physics, Maths, and SNAI. The forms are super quick and will directly impact whether we get extra doubt sessions or adjustments to the teaching pace.

Forms:

SNAI-docs.google.com/forms/…/viewform?usp=sharing&ouid=107954226698023543534…

PSP-docs.google.com/forms/…/viewform?usp=sharing&ouid=107954226698023543534…

Maths-docs.google.com/forms/…/viewform?usp=sharing&ouid=107954226698023543534…

Physics-docs.google.com/forms/…/viewform?usp=sharing&ouid=107954226698023543534…

Appreciate you taking a minute to get this done!

LINKS:
https://docs.google.com/forms/d/e/1FAIpQLSciBSKt9AZAH3gDIoaEGhDbxYBL2SZjNYg2A7Upp4LmDamTkg/viewform?usp=sharing&ouid=107954226698023543534
https://docs.google.com/forms/d/e/1FAIpQLSeCbcCiTZs0AaHHaAjBhAC4i8KshqNVd8b8wKBdrAX9YqkWug/viewform?usp=sharing&ouid=107954226698023543534 https://docs.google.com/forms/d/e/1FAIpQLSdUc-qAxKpDRJbMTIhUtKWhbtvyI8ZoXTr-VknHRBL0Phc-NQ/viewform?usp=sharing&ouid=107954226698023543534
https://docs.google.com/forms/d/e/1FAIpQLSfHmZi4t3PCb1i_GwJGTaOFPJhm2BjD0m7Ej0FbwSfWqIYtgw/viewform?usp=sharing&ouid=107954226698023543534

#academic-announcement-26

nikhil.menon

academic-announcement-26

9 Oct at 2:17 PM

@channel

Applied Physics Syllabus for contest 2 (16/10/2026)

Forces & Types

Newton's Laws of Motion

Frames of Reference

Pseudo Forces & Inertia

Free Body Diagrams

Friction & Types

Normal & Coefficient of Friction

Multi-D FBDs with Friction

Multi-D NLM & Friction Problems

Angular Coordinate System

Radial and Tangential Vectors

CM: Angular Displacement & Speed

CM: Angular Velocity & Acceleration

CM: Centripetal & Centrifugal

Solving Circular Motion

Kinetic Energy

Potential Energy

Work Done

Spring Force

Conservative vs Non-Conservative Forces

Work and Energy Theorem

Centre of Mass

CoM: System of Objects

CoM: Distributed Systems

CoM: Multi-Dimensional Objects

Linear Momentum

Conservation of Linear Momemtum

Conservation of LM: Multi-Dimensional

Conservation of LM: Multi-Stage

Collisions for Rigid Bodies

Elastic Collision

Inelastic Collision

Coefficient of Restitution

Multi-Dimensional Collisions

i.e From Worksheet 05 - 10

#academic-announcement-26

nikhil.menon

academic-announcement-26

14 Oct at 9:55 PM

@channel

Hi everyone,

Please note the following classroom changes for tomorrow:

PSP Lecture: Classroom 8 → Classroom 9

M1 Lecture: Classroom 8 → Classroom 9

S&AI Lecture: Classroom 8 → Classroom 9

Yoga Lecture for Batch B1: Classroom 9

Yoga Lecture for Batch B2: Classroom 5

Students attending PSP, M1, and S&AI lectures are requested to report to Classroom 9. Students from Batch B2 attending Yoga should report to Classroom 5.

Please take note of the changes and proceed to the respective classrooms. (edited)

#academic-announcement-26

Dhairya Agrawal

academic-announcement-26

17 Oct at 11:12 AM

Good morning Sir/Ma'am

I hope you are having a good day. Could you please share the finalized IA1 syllabus and paper pattern for all 5 subjects (PSP, S&AI, AP, Maths, and English) at your earliest convenience? We would be very grateful.

Thank you!

5

#academic-announcement-26

nikhil.menon

academic-announcement-26

19 Oct at 4:35 PM

@channel

Reminder: Math-1 Contest 2

Students, your Math-1 Contest 2 is scheduled for Friday, 23rd October 2026.

Reporting Time: 8:30 AM

Contest Time: 9:00 AM – 11:30 AM

Platform: Product

Math-1 Contest 2 Syllabus

Chapter 1: Algebraic Expressions

Chapter 2: Quadratic Equations

Chapter 3: Inequalities

Chapter 4: Logarithms and Exponents

Chapter 5: Divisibility, GCD/HCF, and LCM

Chapter 6: Congruence, Modular Arithmetic, and the Euclidean Algorithm

Problem-Solving Sheets: 1 and 2

Please prepare well and revise all the topics and problem-solving sheets listed above. Be on time and report by 8:30 AM. All the best!

#academic-announcement-26

nikhil.menon

academic-announcement-26

19 Oct at 4:35 PM

@channel

Reminder: Math-1 IA-1

Students, your Math-1 IA-1 Exam will be a written paper.

MATH IA-1 Syllabus:

Chapter 1: Algebraic Expressions

Chapter 2: Quadratic Equations

Chapter 3: Inequalities

Chapter 4: Logarithms and Exponents

Chapter 5: Divisibility, GCD/HCF, and LCM

Chapter 6: Congruence, Modular Arithmetic, and the Euclidean Algorithm

Problem-Solving Sheets: 1 and 2

(edited)

#academic-announcement-26

nikhil.menon

academic-announcement-26

Yesterday at 6:49 PM

@channel

REMINDER

Students,

Tomorrow, 23rd October 2026, is your M1 Contest 2.

Contest Time: 9:00 AM to 11:00 AM

Reporting Time: 8:30 AM

Contest will be conducted in classroom on Athena

Seating Arrangement: Will be shared by 8:00 AM tomorrow.

All students are requested to go through the exam guidelines carefully before the contest and ensure that you are familiar with all the instructions.

Please report on time and follow the exam guidelines strictly.

All the best!

Exam Guidelines for Student's (1).pdf

Shared by nikhil.menon yesterday

LINKS: https://newtonschoolo-e998430.slack.com/files/U0BQJ0LCF2N/F0CB8U0TSUE/exam_guidelines_for_student_s__1_.pdf?origin_team=T0BNJRD9QT1 https://files.slack.com/files-pri/T0BNJRD9QT1-F0CB8U0TSUE/download/exam_guidelines_for_student_s__1_.pdf?origin_team=T0BNJRD9QT1

#general-announcements-26

nikhil.menon

general-announcements-26

30 Aug at 2:30 PM

@channel

Students,

M1 Lab stands CANCELLED for Batch B1 and B2.

Kindly take note of this.

6

#general-announcements-26

nikhil.menon

general-announcements-26

31 Aug at 7:54 PM

@channel

Students,

Kindly join the respective Slack channels for each subject based on your assigned batch.

Please do not join channels of other batches. Make sure you join all subject channels corresponding to the batch you are enrolled in.

Kindly complete this at the earliest.

#m1-26 #m1-a1 #m1-a2 #m1-b1 #m1-b2

#snai-26 #snai-a1 #snai-a2 #snai-b1 #snai-b2

#psp-26 #psp-a1 #psp-a2 #psp-b1 #psp-b2

#physics-26 #physics-a1 #physics-a2 #physics-b1 #physics-b2

#general-announcements-26

nikhil.menon

general-announcements-26

3 Sep at 6:47 PM

@channel

Students,

Tomorrow's English class is cancelled for both batches A and B.

Revised LHL slots :

Batch A - 2:00 PM to 3:00 PM (Classroom 8)

Batch B - 3:00 PM to 4:00 PM (Classroom 8)

Batch B will have an EXTRA class for M1 from 4:00 PM to 5:00 PM. (Classroom 8) (edited)

#general-announcements-26

nikhil.menon

general-announcements-26

3 Sep at 10:56 PM

@channel

UPDATE

Students,

Revised LHL slots :

Batch A - 2:00 PM to 3:00 PM (Classroom 8)

Batch B - 3:15 PM to 4:15 PM (Classroom 8)

Batch B will have an EXTRA class for M1 from 4:15 PM to 5:15 PM. (Classroom 8)

#general-announcements-26

nikhil.menon

general-announcements-26

7 Sep at 3:20 PM

@channel

Applied Physics Syllabus for contest 1

Sizes in the Universe

Introduction to Measurement

Units and Dimensions

Order of Magnitude

Significant Digits

Scientific Notation

Error and Propagation

Scalars and Vectors

Vector Arithmetic

Vector Products

Vector Resolution

Differentiation

Integration

Distance and Displacement

Speed and Velocity

Acceleration and Jerk

Graphical Understanding of Motion

Motion in Straight Line

Equations of Motion

Relative Motion

Solving 1-D Motion

Projectile Motion

Breaking Down Multi-D Motion

i.e first four lectures (refer provided worksheets for reference and also practice worksheets questions), last 4 topics will be covered in your Wed's lecture be prepared with rest of the topics.

#general-announcements-26

Payal Jain

general-announcements-26

7 Sep at 7:43 PM

Hi @channel,

I’m sharing a Learning Tracker that covers topics across all four subjects (Math, PSP, S&AI, and Applied Physics).

This tracker will help you keep track of what you’ve completed, identify areas that need revision or more practice, and plan your learning consistently.

Please make a copy of each tracker for yourself and update it daily. Once you’ve updated it, share your tracker with me over Slack so I can keep track of your progress and support you wherever needed.

Here are the trackers:

S&AI: docs.google.com/spreadsheets/d/18zDsWPINbLh…/edit?usp=sharing

Mathematics 1: docs.google.com/spreadsheets/d/1pQz1BB1SDS8…/edit?gid=182933272#gid=182933272

Applied Physics: docs.google.com/spreadsheets/d/1ZOx2WQ11_7d…/edit?usp=sharing

PSP: I’ll share the PSP tracker in the next 2–3 days.

Please make sure you start using the available trackers from today itself.

LINKS: https://docs.google.com/spreadsheets/d/18zDsWPINbLhKLMVs8S13rtJQAFUp---_QMasaiwQqzI/edit?usp=sharing https://docs.google.com/spreadsheets/d/1pQz1BB1SDS8NPHzR5RHyG-Z_631Rb40ra2dl8aB0lxo/edit?gid=182933272#gid=182933272 https://docs.google.com/spreadsheets/d/1ZOx2WQ11_7dymZ7tux25VkwUmWyXea1pGLauEx-WNIU/edit?usp=sharing

#general-announcements-26

Payal Jain

general-announcements-26

8 Sep at 3:04 PM

@channel Here are the Learning trackers that covers topics across all four subjects:

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PSP: docs.google.com/spreadsheets/d/1dSKXZJ_aOez…/edit?gid=1592514245#gid=1592514245

Once you've made a copy of all the trackers, please react to this message with a

.

3

LINKS: https://docs.google.com/spreadsheets/d/18zDsWPINbLhKLMVs8S13rtJQAFUp---_QMasaiwQqzI/edit?usp=sharing https://docs.google.com/spreadsheets/d/1pQz1BB1SDS8NPHzR5RHyG-Z_631Rb40ra2dl8aB0lxo/edit?gid=182933272#gid=182933272 https://docs.google.com/spreadsheets/d/1ZOx2WQ11_7dymZ7tux25VkwUmWyXea1pGLauEx-WNIU/edit?usp=sharing https://docs.google.com/spreadsheets/d/1dSKXZJ_aOezRekJ1RqTcPjQDa-AsB85xYugHMpaYYgA/edit?gid=1592514245#gid=1592514245

#general-announcements-26

nikhil.menon

general-announcements-26

10 Sep at 8:11 PM

@channel

Students,

Batch B will have an extra class for M1 tomorrow from 3:00 PM to 4:30PM in classroom 8.

Kindly take a note of the same. (edited)

#general-announcements-26

nikhil.menon

general-announcements-26

11 Sep at 2:12 PM

@channel

Students,

A mandatory Math lecture is scheduled for Batch B from 5:00 PM to 6:30 PM in classroom 8. (edited)

#general-announcements-26

nikhil.menon

general-announcements-26

11 Sep at 4:45 PM

@channel

REMINDER

Students,

On 18th September 2026, is your Math-1 Contest 1.

Contest Time: 9:00 AM to 11:00 AM

Reporting Time: 8:30 AM

Course Syllabus: Chapters 1–3

Chapter 1: Algebraic Expressions

Chapter 2: Quadratic Equations

Chapter 3: Inequalities

#student-progression-batch-a

Kartik Khandelwal

student-progression-batch-a

25 Aug at 1:43 PM

mathsisfun.com/games/broken-calculator.html

Broken Calculator

Play Broken Calculator. Sam's calculator got smashed, but some of it still works. Can you help Sam get the answers?

Math is Fun

LINKS: https://www.mathsisfun.com/games/broken-calculator.html

#student-progression-batch-a

Payal Jain

student-progression-batch-a

7 Sep at 7:42 PM

Hi @channel,

I’m sharing a Learning Tracker that covers topics across all four subjects (Math, PSP, S&AI, and Applied Physics).

This tracker will help you keep track of what you’ve completed, identify areas that need revision or more practice, and plan your learning consistently.

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#student-progression-batch-a

Payal Jain

student-progression-batch-a

8 Sep at 3:04 PM

@channel Here are the Learning trackers that covers topics across all four subjects:

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PSP: docs.google.com/spreadsheets/d/1dSKXZJ_aOez…/edit?gid=1592514245#gid=1592514245

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LINKS: https://docs.google.com/spreadsheets/d/18zDsWPINbLhKLMVs8S13rtJQAFUp---_QMasaiwQqzI/edit?usp=sharing https://docs.google.com/spreadsheets/d/1pQz1BB1SDS8NPHzR5RHyG-Z_631Rb40ra2dl8aB0lxo/edit?gid=182933272#gid=182933272 https://docs.google.com/spreadsheets/d/1ZOx2WQ11_7dymZ7tux25VkwUmWyXea1pGLauEx-WNIU/edit?usp=sharing https://docs.google.com/spreadsheets/d/1dSKXZJ_aOezRekJ1RqTcPjQDa-AsB85xYugHMpaYYgA/edit?gid=1592514245#gid=1592514245

#student-progression-batch-b

Payal Jain

student-progression-batch-b

7 Sep at 7:42 PM

Hi @channel,

I’m sharing a Learning Tracker that covers topics across all four subjects (Math, PSP, S&AI, and Applied Physics).

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```
#student-progression-batch-b

Payal Jain



student-progression-batch-b

8 Sep at 3:04 PM

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Applied Physics: docs.google.com/spreadsheets/d/1ZOx2WQ11_7d…/edit?usp=sharing

PSP: docs.google.com/spreadsheets/d/1dSKXZJ_aOez…/edit?gid=1592514245#gid=1592514245

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.

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usp=sharing https://docs.google.com/spreadsheets/d/1pQz1BB1SDS8NPHzR5RHyG-
Z_631Rb40ra2dl8aB0lxo/edit?gid=182933272#gid=182933272
https://docs.google.com/spreadsheets/d/1ZOx2WQ11_7dymZ7tux25VkwUmWyXea1pGLauEx-WNIU/edit?
usp=sharing https://docs.google.com/spreadsheets/d/1dSKXZJ_aOezRekJ1RqTcPjQDa-
AsB85xYugHMpaYYgA/edit?gid=1592514245#gid=1592514245
```

WHERE THESE QUESTIONS CAME FROM (sources)

- **Subject assignment pages / My Assignments + Assignment Contests API** — 0 assignments, 0 questions; full statement captured from each question's playground page (read-only visit) — 0/0 texts present.
- **Lecture pages** (Course Content sidebar list) — 0 of 32 lecture windows carry content: post-class question text below the video (`Read More` expanded) and in-class quiz questions via read-only views only (Revise / View Solutions / Next-Previous walk — never Attempt/Start/Submit).
- **Assessments API** — 0 quiz schedules (title, window, XP earned). No question text in the API (hidden until test start).
- **Slack search harvest** (`slack_harvest.py`, 2026-10-23) — 792 messages across #m1-26/#m1-b1 plus math blocks of #academic-announcement-26/#general-announcements-26/#student-progression-batch-a/b (read-only search; no Send-button use): official Contest 1/2 + IA-1 syllabi, the Contest 1 paper's full text, the learning tracker's 172 topic rows.

NOT FETCHED — open these yourself

(These sit behind `Attempt Now` / `Start Test` buttons. I never click those — they consume quiz attempts. Everything else above is complete.)

Quiz Contests tab (separate MCQ tests)

- Quiz contest questions only appear after `Start Test` (graded, consumes an attempt): open the Quiz Contests tab manually if you want these: the Math-1 Quiz Contest listed on the course Quiz Contests tab
- What IS captured: all assignment questions (0), all in-class quiz questions shown above, all contest ASSIGNMENT texts

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TOTALS: 0 assignment questions (0 solved), 0 quizzes, 0 questions without captured text, 0 lecture windows with content, Slack: 792 msgs, 172 tracker topics